

Temperature Scaling

Until now, the generated token is selected corresponding to the largest probability score among all tokens in the vocab
↳ Leads to a lot of randomness and diversity in generated text
↳ We will learn 2 techniques to control this randomness

↓
Temperature
Scaling

↓
Top-k
Sampling

Technique 1: Temperature Scaling

↳ Replace softmax w/ probability distribution

○ Probability Distribution → Multinomial probability distribution
Samples next token according to probability score

* What is temperature? Fancy term for dividing the logits by a # greater than zero

Scaled logits = $\frac{\text{logits}}{\text{Temperature}}$

small: Sharper distribution

Large: Blottered distribution

more variety, but also more nonsense)

↓
Softmax

↓
Tensor of
Probabilities

logits → Temperature → Softmax

↓
Multinomial
Sampling

○ As the temperature becomes high,
its like the entropy increases

↳ Every next token has some
probability of being the next token

Whereas if the temperature is really
low, the entropy is very low.

↳ It's like everything is concentrated
to one single token, and we have
a sharper probability distribution